

where:

$$\begin{aligned}
m_q &= \left(\frac{\pi}{1+\delta} \boldsymbol{\mu}^\top + \frac{\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \right) \bar{\mathbf{Q}} \boldsymbol{\mu}, \\
\nu_q &= \frac{\pi^2}{(1+\delta)^2} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} + \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta + \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta \\
&\quad + \frac{\pi T}{(1+\delta)^2} \left(1 - \frac{2\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} - \frac{2\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \\
&\quad + \frac{(1-\pi)\sigma^2 T}{(1+\delta_S)^2} \left(\alpha - \frac{2\lambda^2(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta - \frac{2\lambda\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right).
\end{aligned}$$

With:

$$\lambda = \phi(1 - \varepsilon) - \rho\varepsilon, \quad \delta_S = \alpha\sigma^2\delta$$

D RANDOM MATRIX ANALYSIS OF DISTRIBUTION SHIFT

We will now quantify the performance of the classifier obtained through mixing some real data and synthetic data sampled according to the schema described in 2. Hence, the matrix $\bar{\mathbf{Q}}$, defined in lemma A.4, is no longer deterministic as we take the covariance matrix $\hat{\mathbf{C}} = \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} (\mathbf{x}_i - y_i \hat{\boldsymbol{\mu}})(\mathbf{x}_i - y_i \hat{\boldsymbol{\mu}})^\top$. For simplicity, and without loss of generality, we consider $\hat{n}$ Gaussian vectors $(\mathbf{z}_i)_{i=1}^{\hat{n}} \sim \mathcal{N}(0, \mathbf{I}_p)$ that are independent of $(\mathbf{x}_i)_{i=1}^{\hat{n}}$, and write:

$$\hat{\boldsymbol{\mu}} = \boldsymbol{\mu}_\beta = \boldsymbol{\mu} + \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbf{z}_i, \quad \hat{\mathbf{C}} = \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbf{z}_i \mathbf{z}_i^\top$$

Note that we can ignore the error of estimation of $\hat{\boldsymbol{\mu}}$ because we have that:

$$\mathbb{E} \left[\frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbf{z}_i \right] = 0, \quad \mathbb{E} \left[\left(\frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbf{z}_i \right) \left(\frac{1}{\hat{n}} \sum_{j=1}^{\hat{n}} \mathbf{z}_j \right)^\top \right] = \frac{1}{\hat{n}} \mathbf{I}_p$$

Hence, when we have a sufficiently large $\hat{n}$, we will assume that: $\hat{\boldsymbol{\mu}} = \boldsymbol{\mu}$ (the estimation error is on $\mathcal{O}(\hat{n}^{-1})$).

D.1 DETERMINISTIC EQUIVALENTS:

The resolvent matrix to be considered in this setting is the one defined in lemma A.4 but with $\hat{\mathbf{C}}$:

$$\begin{aligned} \bar{\mathbf{Q}}(\gamma) &= \left(\left(\frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} \right) \boldsymbol{\mu} \boldsymbol{\mu}^\top + \frac{\alpha(1-\pi)}{1+\delta_S} \hat{\mathbf{C}} + \left(\gamma + \frac{\pi}{1+\delta} \right) \mathbf{I}_p \right)^{-1} \\ &= \left(\left(\frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} \right) \boldsymbol{\mu} \boldsymbol{\mu}^\top + \frac{\alpha(1-\pi)}{(1+\delta_S)\hat{n}} \sum_{i=1}^{\hat{n}} \mathbf{z}_i \mathbf{z}_i^\top + \left(\gamma + \frac{\pi}{1+\delta} \right) \mathbf{I}_p \right)^{-1} \end{aligned}$$

where:

$$\delta = \frac{1}{N} \text{Tr}(\bar{\mathbf{Q}}), \quad \delta_S = \frac{\alpha}{N} \text{Tr}(\hat{\mathbf{C}} \bar{\mathbf{Q}})$$

Let us denote by $\bar{\mathbf{Q}}_{-i}$ the resolvent matrix gotten by removing its dependence on the vector $\mathbf{z}_i$. In other words:

$$\bar{\mathbf{Q}}_{-i} = \left(\bar{\mathbf{Q}} - \frac{\alpha(1-\pi)}{\hat{n}(1+\delta_S)} \mathbf{z}_i \mathbf{z}_i^\top \right)^{-1}, \quad \bar{\mathbf{Q}} = \left(\bar{\mathbf{Q}}_{-i} + \frac{\alpha(1-\pi)}{\hat{n}(1+\delta_S)} \mathbf{z}_i \mathbf{z}_i^\top \right)^{-1}$$

By Sherman-Morrisson's lemma A.3, we have that:

$$\bar{\mathbf{Q}} = \bar{\mathbf{Q}}_{-i} - \frac{\frac{\alpha(1-\pi)}{\hat{n}(1+\delta_S)} \bar{\mathbf{Q}}_{-i} \mathbf{z}_i \mathbf{z}_i^\top \bar{\mathbf{Q}}_{-i}}{1 + \frac{\alpha(1-\pi)}{\hat{n}(1+\delta_S)} \mathbf{z}_i^\top \bar{\mathbf{Q}}_{-i} \mathbf{z}_i}$$

And:

$$\bar{\mathbf{Q}} \mathbf{z}_i = \frac{\bar{\mathbf{Q}}_{-i} \mathbf{z}_i}{1 + \frac{\alpha(1-\pi)}{\hat{n}(1+\delta_S)} \mathbf{z}_i^\top \bar{\mathbf{Q}}_{-i} \mathbf{z}_i} = \frac{\bar{\mathbf{Q}}_{-i} \mathbf{z}_i}{1 + \bar{\delta}}$$

where:

$$\bar{\delta} = \frac{\alpha(1-\pi)}{1+\delta_S} \frac{1}{\hat{n}} \text{Tr}(\bar{\mathbf{Q}}) \quad (27)$$

Since the covariance estimate in equation 2 is stochastic, the matrix $\bar{\mathbf{Q}}$ is no longer deterministic when replacing $\mathbf{C}$ with $\hat{\mathbf{C}}$. Hence, we will give a further deterministic equivalent to $\bar{\mathbf{Q}}$ in the following lemma.

Lemma D.1 (Second Deterministic equivalent). *A deterministic equivalent of $\bar{\mathbf{Q}}$ is given by:*

$$\bar{\mathbf{Q}} = \left(\left(\frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} \right) \boldsymbol{\mu} \boldsymbol{\mu}^\top + \left(\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{(1+\delta_S)(1+\bar{\delta})} \right) \mathbf{I}_p \right)^{-1}$$

where $\bar{\delta}$ can be found as a fixed point using the following identity:

$$\bar{\delta} = \frac{\alpha(1-\pi)}{(1+\delta_S)} \frac{1}{\hat{n}} \text{Tr}(\bar{\mathbf{Q}}) = \frac{\alpha(1-\pi)}{(1+\delta_S)} \frac{\frac{\hat{n}}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{(1+\delta_S)(1+\bar{\delta})}}}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{(1+\delta_S)(1+\bar{\delta})}}, \quad \delta = \frac{1}{N} \text{Tr}(\bar{\mathbf{Q}}) = \frac{\hat{n}}{N} \frac{(1+\delta_S)}{\alpha(1-\pi)} \bar{\delta}$$

$$\delta_S = \frac{\alpha}{N} \text{Tr}(\mathbb{E}[\hat{\mathbf{C}}\bar{\mathbf{Q}}]) = \frac{\alpha\bar{\delta}}{1+\bar{\delta}}$$

Now we will prove the deterministic equivalent given by lemma D.1.

PROOF OF LEMMA D.1:

Let us denote $\bar{\mathbf{Q}} = \left(\frac{\alpha(1-\pi)}{1+\delta_S} \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbf{z}_i \mathbf{z}_i^\top + \mathbf{A} \right)^{-1}$. Let also $\bar{\bar{\mathbf{Q}}}$ be the deterministic equivalent of $\bar{\mathbf{Q}}$. It can be written as: $\bar{\mathbf{Q}} = (\mathbf{S} + \mathbf{A})^{-1}$. We want to find some $\mathbf{S}$ such that for all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^p$:

$$\mathbf{a}^\top \mathbb{E}[\bar{\mathbf{Q}}] \mathbf{b} \rightarrow \mathbf{a}^\top \bar{\bar{\mathbf{Q}}} \mathbf{b}$$

We have that:

$$\begin{aligned} \mathbb{E}[\bar{\mathbf{Q}}] - \bar{\bar{\mathbf{Q}}} &= \mathbb{E} \left[\bar{\mathbf{Q}} \left(\mathbf{S} - \frac{\alpha(1-\pi)}{1+\delta_S} \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbf{z}_i \mathbf{z}_i^\top \right) \bar{\bar{\mathbf{Q}}} \right] \\ &= \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbb{E} \left[\bar{\mathbf{Q}} \left(\mathbf{S} - \frac{\alpha(1-\pi)}{1+\delta_S} \mathbf{z}_i \mathbf{z}_i^\top \right) \bar{\bar{\mathbf{Q}}} \right] \\ &= \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbb{E} \left[\bar{\mathbf{Q}} \mathbf{S} - \frac{\alpha(1-\pi)}{1+\delta_S} \frac{1}{1+\bar{\delta}} \bar{\mathbf{Q}}_{-i} \mathbf{z}_i \mathbf{z}_i^\top \right] \bar{\bar{\mathbf{Q}}} \\ &= \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbb{E} \left[\bar{\mathbf{Q}}_{-i} \left(\mathbf{S} - \frac{\alpha(1-\pi)}{1+\delta_S} \frac{1}{1+\bar{\delta}} \mathbf{z}_i \mathbf{z}_i^\top \right) \right] \bar{\bar{\mathbf{Q}}} + \mathcal{O}(\hat{n}^{-1}) \end{aligned}$$

Hence, it suffices to have $\mathbf{S} = \mathbb{E} \left[\frac{\alpha(1-\pi)}{1+\delta_S} \frac{1}{1+\bar{\delta}} \mathbf{z}_i \mathbf{z}_i^\top \right] = \frac{\alpha(1-\pi)}{1+\delta_S} \frac{1}{1+\bar{\delta}} \mathbf{I}_p$, and thus:

D.2 DETERMINISTIC EQUIVALENT OF $\bar{\mathbf{Q}} \mathbf{A} \bar{\mathbf{Q}}$:

Let $\mathbf{A} \in \mathbb{R}^{p \times p}$ be some deterministic matrix. We have that:

$$\begin{aligned} \mathbb{E}[\bar{\mathbf{Q}} \mathbf{A} \bar{\mathbf{Q}}] &= \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\bar{\mathbf{Q}}} + \mathbb{E}[(\bar{\mathbf{Q}} - \bar{\bar{\mathbf{Q}}}) \mathbf{A} \bar{\mathbf{Q}}] \\ &= \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\bar{\mathbf{Q}}} + \mathbb{E}[\bar{\mathbf{Q}} (\bar{\bar{\mathbf{Q}}}^{-1} - \bar{\mathbf{Q}}^{-1}) \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}}] \\ &= \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\bar{\mathbf{Q}}} + \frac{\alpha(1-\pi)}{(1+\delta_S)} \mathbb{E} \left[\bar{\mathbf{Q}} \left(\frac{1}{1+\bar{\delta}} - \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbf{z}_i \mathbf{z}_i^\top \right) \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}} \right] \\ &= \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\bar{\mathbf{Q}}} + \frac{\alpha(1-\pi)}{(1+\delta_S)} \left(\frac{1}{1+\bar{\delta}} \mathbb{E}[\bar{\mathbf{Q}} \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}}] - \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} \mathbb{E}[\bar{\mathbf{Q}} \mathbf{z}_i \mathbf{z}_i^\top \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}}] \right) \end{aligned}$$

And we have that for $i \in \{1, \dots, \hat{n}\}$:

$$\begin{aligned} \mathbb{E}[\bar{\mathbf{Q}} \mathbf{z}_i \mathbf{z}_i^\top \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}}] &= \frac{1}{1+\bar{\delta}} \mathbb{E}[\bar{\mathbf{Q}}_{-i} \mathbf{z}_i \mathbf{z}_i^\top \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}}] \\ &= \frac{1}{1+\bar{\delta}} \mathbb{E} \left[\bar{\mathbf{Q}}_{-i} \mathbf{z}_i \mathbf{z}_i^\top \bar{\bar{\mathbf{Q}}} \mathbf{A} \left(\bar{\mathbf{Q}}_{-i} - \frac{\alpha(1-\pi)}{\hat{n}(1+\delta_S)(1+\bar{\delta})} \bar{\mathbf{Q}}_{-i} \mathbf{z}_i \mathbf{z}_i^\top \bar{\mathbf{Q}}_{-i} \right) \right] \\ &= \frac{1}{1+\bar{\delta}} \mathbb{E}[\bar{\mathbf{Q}}_{-i} \mathbf{z}_i \mathbf{z}_i^\top \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}}_{-i}] - \frac{\alpha(1-\pi)}{\hat{n}(1+\delta_S)(1+\bar{\delta})^2} \mathbb{E}[\bar{\mathbf{Q}}_{-i} \mathbf{z}_i \mathbf{z}_i^\top \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}}_{-i} \mathbf{z}_i \mathbf{z}_i^\top \bar{\mathbf{Q}}_{-i}] \\ &= \frac{1}{1+\bar{\delta}} \mathbb{E}[\bar{\mathbf{Q}} \bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\mathbf{Q}}] - \frac{\alpha(1-\pi)}{\hat{n}(1+\delta_S)(1+\bar{\delta})^2} \text{Tr}(\bar{\bar{\mathbf{Q}}} \mathbf{A} \bar{\bar{\mathbf{Q}}}) \mathbb{E}[\bar{\mathbf{Q}}^2] \end{aligned}$$